

Homework 4 (of 5)

Probability and Statistics - solution

5.16 A survey conducted by a song rating service finds that the percentages of listeners familiar with *Poker Face* by Lady Gaga who would give the song ratings of 5, 4, 3, 2, and 1 are, respectively, 43 percent, 21 percent, 22 percent, 7 percent, and 7 percent. Assign the numerical values 1, 2, 3, 4, and 5 to the (qualitative) ratings 1, 2, 3, 4, and 5 and find an estimate of the probability distribution of x this song's rating by a randomly selected listener who is familiar with the song.

Solution

Rate, x	Probability, $p(x)$
1	$f(5) = 0.43$
2	$f(4) = 0.21$
3	$f(3) = 0.22$
4	$f(2) = 0.07$
5	$f(1) = 0.07$

5.17 In Exercise 5.16,

- (a) Find the **expected value** of the estimated probability distribution.
- (b) Explain the meaning of this expected value in terms of all possible *Poker Face* listeners.

Solution

$$\begin{aligned}\mu_x &= 5 \times 0.43 + 4 \times 0.21 + 3 \times 0.22 + 2 \\ &\times 0.07 + 1 \times 0.07 = 3.86\end{aligned}$$

By average, all possible *Poker Face* listeners give the song rating of 3.86

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5.23

The customer service department for a wholesale electronics outlet claims that 90 percent of all customer complaints are resolved to the satisfaction of the customer. In order to test this claim, a random sample of 15 customers who have filed complaints is selected.

- a** Let x = the number of sampled customers whose complaints were resolved to the customer's satisfaction. Assuming the claim is true, write the binomial formula for this situation.
- b** Use the binomial tables (see Table A.1, page 783) to find each of the following if we assume that the claim is true:
 - (1) $P(x \leq 13)$.
 - (2) $P(x > 10)$.
 - (3) $P(x \geq 14)$.
 - (4) $P(9 \leq x \leq 12)$.
 - (5) $P(x \leq 9)$.
- c** Suppose that of the 15 customers selected, 9 have had their complaints resolved satisfactorily. Using part *b*, do you believe the claim of 90 percent satisfaction? Explain.

Answer

$$P(x) = \frac{15!}{x! (15 - x)!} 0.9^x 0.1^{15-x}$$

(b)

$$P(x \leq 13) = 0.2669 + 0.1285 + 0.0428 + 0.0105 + 0.0019 + 0.0003 = 0.4509$$

$$P(x > 10) = 0.9873$$

$$P(x \geq 14) = 0.3432 + 0.2059 = 0.5491$$

$$P(9 \leq x \leq 12) = 0.1837$$

$$P(x \leq 9) = 0.0022$$

(c) Suppose that of the 15 customers selected, 9 have had their complaints resolved satisfactorily. Using part b, do you believe the claim of 90 percent satisfaction? Explain. (No, $P(x \leq 9)$ is too small)

$n = 15$ p $x \downarrow$ **.05****.10****.15****.20****.25****.30****.35****.40****.45****.50**

0

.4633

.2059

.0874

.0352

.0134

.0047

.0016

.0005

.0001

.0000

15

1

.3658

.3432

.2312

.1319

.0668

.0305

.0126

.0047

.0016

.0005

14

2

.1348

.2669

.2856

.2309

.1559

.0916

.0476

.0219

.0090

.0032

13

3

.0307

.1285

.2184

.2501

.2252

.1700

.1110

.0634

.0318

.0139

12

4

.0049

.0428

.1156

.1876

.2252

.2186

.1792

.1268

.0780

.0417

11

5

.0006

.0105

.0449

.1032

.1651

.2061

.2123

.1859

.1404

.0916

10

6

.0000

.0019

.0132

.0430

.0917

.1472

.1906

.2066

.1914

.1527

9

7

.0000

.0003

.0030

.0138

.0393

.0811

.1319

.1771

.2013

.1964

8

8

.0000

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.0005

.0035

.0131

.0348

.0710

.1181

.1647

.1964

7

9

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4

12

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0

.95**.90****.85****.80****.75****.70****.65****.60****.55****.50** $x \uparrow$

5.27

The January 1986 mission of the Space Shuttle *Challenger* was the 25th such shuttle mission. It was unsuccessful due to an explosion caused by an O-ring seal failure.

- a** According to NASA, the probability of such a failure in a single mission was $1/60,000$. Using this value of p and assuming all missions are independent, calculate the probability of no mission failures in 25 attempts. Then calculate the probability of at least one mission failure in 25 attempts.
- b** According to a study conducted for the Air Force, the probability of such a failure in a single mission was $1/35$. Recalculate the probability of no mission failures in 25 attempts and the probability of at least one mission failure in 25 attempts.
- c** Based on your answers to parts *a* and *b*, which value of p seems more likely to be true? Explain.
- d** How small must p be made in order to ensure that the probability of no mission failures in 25 attempts is .999?

Answer

(a)

$$P(x) = \frac{25!}{x! (25 - x)!} \left(\frac{1}{60,000} \right)^x \left(1 - \frac{1}{60,000} \right)^{25-x}$$

$$P(x = 0) = 0.99958$$

$$P(x \geq 1) = 1 - P(x = 0) = 0.0004166$$

(b)

$$P(x) = \frac{25!}{x! (25 - x)!} \left(\frac{1}{35} \right)^x \left(1 - \frac{1}{35} \right)^{25-x}$$

$$P(x = 0) = 0.4845$$

$$P(x \geq 1) = 1 - P(x = 0) = 0.5155$$

(c) The probability (b) is more likely to be true

(d)

$$P(x) = \frac{25!}{x! (25 - x)!} (p)^x (1 - p)^{25-x}$$

$$(1 - p)^{25} = 0.999$$

$$(1 - p)^{25} = 0.999$$

$$p = 0.000040019 = \frac{1}{24988}$$